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MAJOR ARTICLE

Reproducible identification of *Staphylococcus aureus* bacteremia clinical subphenotypes

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Background: Clinical heterogeneity in *Staphylococcus aureus* bacteremia (SAB) complicates clinical management and research. We have previously identified five clinically distinct subphenotypes of SAB associated with differences in outcomes and response to adjunctive rifampicin. Here, we aimed to identify these subphenotypes in geographically diverse

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observational cohorts, including a higher prevalence of methicillin-resistant *S. aureus* (MRSA) bacteremia and the USA300 clone.

Methods: We studied three cohorts of adults with SAB from observational studies: a UK retrospective study (Edinburgh cohort 2, n=463); a Dutch prospective study (IDISA, n=490); and a USA prospective study (SABG-PCS, n=755). Subphenotypes were identified from routinely available clinical data using latent class analysis.

Results: Patients from the SABG-PCS cohort had greater multimorbidity and more MRSA bacteremia (40.2%, 303/755), including infection with the USA300 clone (14.7%, 111/755). Five distinct subphenotypes were identified in each cohort: (A) older age and cardio-metabolic multimorbidity; (B) nosocomial acquisition and intravenous catheter portal of entry; (C) community acquisition and metastatic infection; (D) chronic kidney disease; and (E) younger age, injection drug use, and metastatic infection. Bacterial genotypes varied substantially between the Edinburgh 2 and SABG-PCS cohorts but did not differ between subphenotypes within each cohort. 90-day mortality was highest in subphenotype A, and persistent bacteremia in subphenotypes C and E.

Conclusions: We have reproducibly identified five clinical subphenotypes of SAB in observational cohorts including diverse bacterial genetic lineages and a cohort with a high prevalence of MRSA and USA300 bacteremia. These robustly reproducible clinical subphenotypes provide a framework to rationalize the heterogeneity intrinsic to SAB.

Keywords: *Staphylococcus aureus*; bacteremia; patient stratification; subphenotypes; MRSA

INTRODUCTION

Staphylococcus aureus bacteremia (SAB) is a common and complicated infection, globally accounting for the largest number of deaths associated with bacteremia[1, 2]. In addition to mortality, SAB can be complicated by metastatic foci of infection, recurrence despite seemingly appropriate treatment, and associated morbidity[3, 4]. The heterogeneity intrinsic to SAB challenges efficacy of universal treatment approaches, complicating the management of individual patients in addition to research efforts to improve treatment outcomes[5]. In our previous work, we demonstrated that this heterogeneity can be rationalized using latent class analysis (LCA) to identify five subphenotypes of SAB representing clinically distinct and readily interpretable disease archetypes, formed without consideration of outcome data[6, 7]. Supporting the clinical relevance of this approach, these subphenotypes were associated with differences in mortality and microbiologic outcomes, and differential treatment effects of adjunctive rifampicin in the ARREST trial[8]. These findings were based on three cohorts from the United Kingdom and Spain, two of which were from clinical trials (ARREST and SAFO[8, 9]), including only 64/1430 patients with methicillin-resistant *S. aureus* (MRSA) bacteremia. The generalizability of these findings to ‘real life’ patient cohorts in general, and MRSA bacteremia specifically, remains unresolved. Epidemiologic differences such as the prevalence of MRSA and the hypervirulent USA300 clone differ significantly across regions and are known to influence outcomes[10-12]. To address this, we aimed to replicate the identification of these previously identified subphenotypes using three geographically diverse observational cohorts, including one from the United States of America with a higher prevalence of MRSA bacteremia and the USA300 clone.

METHODS

Patient cohorts

Patients were included from three observational studies: a cohort from an ongoing United Kingdom (Scotland) retrospective study (Edinburgh cohort 2, n=463)[3, 6]; the Dutch IDISA (Improved Diagnostic Strategies in *S. aureus* bacteremia) multicenter prospective study (IDISA cohort, n=490)[13], and the United States of America SABG-PCS (*S. aureus* Bacteremia Group Prospective Cohort Study) prospective study (SABG-PCS cohort, n=755)[11]. The Edinburgh 2 cohort included consecutive unique adults (aged ≥18 years) with monomicrobial SAB between 28 August 2022 and 18 December 2024, approved by the South East Scotland Research Ethics Committee. No patients included in our previous analysis were included in this second cohort of patients. The IDISA cohort included adults (aged ≥18 years) with SAB between 1 July 2017 to 30 September 2019, from seven hospitals in the Randstad metropolitan region of the Netherlands, approved by the Medical Ethics Committee of the Academic Medical Centre Amsterdam. The cohort from SABG-PCS analyzed in this study included adults (aged ≥18 years) with monomicrobial SAB between 1 January 2016 to 1 January 2020, at Duke University Medical Center in North Carolina, approved by the Duke Institutional Review Board.

Variable definitions

The Charlson Comorbidity Index (CCI) was used to define comorbidities, adjusted for age. Vascular disease was defined as a history of peripheral vascular disease, myocardial infarction, or stroke. The route of acquisition of infection was classified based on the criteria by Friedman *et al*[14]. The portal of entry was defined as the most probable entry point of *S. aureus* into the bloodstream. Metastatic infection was defined as the clinical, radiologic, or microbiologic identification of foci of infection remote from the portal of entry, thought to have arisen through hematogenous dissemination, and included infective endocarditis. In the SABG-PCS cohort, heart rate, temperature, and creatinine were collected as ordinal variables (**Table S1**). For the Edinburgh 2 and IDISA cohorts, they were continuous variables. In the IDISA cohort, recorded vital signs and laboratory measurements were the most extreme values within 24 hours of the index blood culture. In the Edinburgh 2 and SABG-PCS cohorts, vital signs and laboratory measurements were recorded closest to the time of the index blood culture. In the Edinburgh 2 cohort, *spa* typing and Panton-Valentine leucocidin (*pvl*) gene qRT-PCR were performed by the Scottish Microbiology

Reference Laboratory. In the SABG-PCS cohort, *spa* typing and confirmation of USA300 genotype were performed as previously described[11]. Persistent bacteremia was defined as a further positive blood culture >48 hours after the index blood culture in the Edinburgh 2 cohort[15], and >96 hours in the IDISA cohort. In the SABG-PCS cohort, persistent bacteremia was defined as persistently positive blood cultures for ≥5 days after initiation of appropriate antimicrobial therapy. Recurrent bacteremia was defined as a further positive blood culture within 90 days of stopping treatment in the Edinburgh 2 and SABG-PCS cohorts, and within 90 days of the index blood culture in the IDISA cohort. In the Edinburgh 2 cohort, this was restricted to bacteremia caused by an isolate from the same *spa*-inferred clonal complex as the index bacteremia.

Statistical analysis

Latent class analysis (LCA) was used to identify subgroups within the different cohorts. A detailed description was given previously[6]. Indicator variables used to identify subgroups were the same as our previous analysis with the exception that endocarditis and other metastatic complications were included as a single variable, rather than separately, due to data availability . Classes were formed without consideration of outcomes. Nonnormally distributed variables with right-skew were log-transformed. Missing values were handled with full information maximum likelihood. A combination of the Bayesian Information criteria (BIC), number of classes, size of the smallest class, and clinical interpretation were used for model selection. We first identified models using the BIC (the most conservative index where a decrease indicates a more informative model), then checked smallest class sizes (to avoid uninformative and underpowered classes), then applied clinical interpretation (*e.g.* to ensure that all classes were clinically distinct). To avoid a local maximum, 16 random starting values were used, with 50 iterations performed for each starting value[16]. The results were reviewed to confirm that the same maximum likelihood solution was found. Following class identification, we calculated the posterior probability of class membership for each individual across all identified classes, and assigned each individual to the class with the highest posterior class membership probability. We corrected for misclassification error using bias-adjusted 3-step LCA[17]. LCA was done using the Latent GOLD 6.1 statistical software package[18].

For comparison of continuous variables between two cohorts, the Mann-Whitney U test was used. For comparison of continuous variables between all three cohorts, the Kruskal-Wallis test was used. Categorical variables were compared by contingency table analysis, using the Chi-squared if cell counts were ≥5 or Fisher's exact test if cell counts were <5. A *P* value of <0.05 was deemed to indicate statistical significance. Z-scores were used to visualize the comparison of variables and outcomes between sub-phenotypes, calculated using the formula $z = \frac{(value\ for\ subphenotype\ - mean\ for\ variable)}{standard\ deviation\ for\ variable}$. GraphPad Prism Version 10.5.0 for macOS was used for data visualisation and analysis.

RESULTS

Cohort characteristics

Characteristics of the Edinburgh 2, IDISA, and SABG-PCS cohorts are compared in **Table 1**. Notable differences between these cohorts included a lower prevalence of chronic kidney disease in the Edinburgh 2 cohort and a lower prevalence of people who inject drugs in the IDISA cohort. In the SABG-PCS cohort there was substantially greater multimorbidity (as determined by the Charlson Comorbidity Index) despite a slightly lower median age. Infective endocarditis was diagnosed less frequently in the Edinburgh 2 cohort. MRSA bacteremia was uncommon in the Edinburgh 2 and IDISA cohorts (2.8% and 2.0% respectively) compared to the high rate in the SABG-PCS cohort (40.2%). Persistent bacteremia was also more common in the SABG-PCS cohort despite a more stringent definition (requiring positive blood cultures for ≥5 days after starting treatment). Across the cohorts, most episodes of SAB were acquired in the community, and an unknown portal of entry was the most frequent individual category, followed by skin or soft tissue infection, and intravenous catheter.

Identification of clinical subphenotypes

Consistent with our previous work, BIC values, class sizes, and clinical interpretability together favored the five class models (**Table S2**). Overall, when class-defining variables were compared between the classes within each cohort, the five classes replicated the previously described subphenotypes A to E (**Figure 1; Tables S3-S5**). Subphenotype A was associated with older age, male sex, and multimorbidity (dementia and vascular disease). Subphenotype B was associated with nosocomial SAB, intravenous catheter as the portal of entry, less metastatic infection, and overall less comorbidity. Subphenotype C was associated with community-acquired SAB from an unknown portal of entry, with metastatic infection and higher C-reactive protein. Subphenotype D was associated with chronic kidney disease (CKD), healthcare contact (either community-acquired healthcare-associated or nosocomial acquisition), and vascular disease. Hemodialysis was not included as a class-defining variable but was common in subphenotype D (Edinburgh 2, n=16/34; IDISA, n=22/92; SABG-PCS, n=76/124). Subphenotype E was associated with SAB in younger people without CKD, dementia or vascular disease. This subphenotype was associated with acquisition through injection drug use, metastatic infection, and co-existent liver disease. Certain characteristics were less consistently associated with clinical subphenotype. Of particular clinical relevance, the presence of cardiac prosthetic material was variably associated with subphenotypes A, C, and D across the three cohorts. The association of subphenotype A with site of acquisition was also inconsistent (predominantly community-onset non-healthcare associated in Edinburgh 2 cohort, community-onset healthcare associated in IDISA, and nosocomial in SABG-PCS).

The contribution of each class-defining variable to the clustering was quantified using the R^2 value (**Figure S1**), indicating that age, acquisition, CKD, injection drug use, metastatic infection, and portal of entry made the greatest individual contributions to class assignment. MRSA bacteremia was consistently associated with subphenotypes A (old age and multimorbidity), B (nosocomial) and E (injection drug use) but overall did not make a large contribution to the clustering.

Comorbidity and multimorbidity

The extent of multimorbidity, determined using the CCI, differed between each of the three cohorts with patients in the SABG-PCS cohort having the highest CCI and patients in the Edinburgh 2 cohort having the lowest (**Table 1; Figure S2**). CCI scores were compared across subphenotypes, identifying the greatest extent of multimorbidity in subphenotypes A and D, and the lowest in B and E, consistent with our previous findings (**Figure 2A**). Despite the greater baseline multimorbidity, this pattern of differences in CCI scores remained in the SABG-PCS cohort. Diabetes mellitus was not included as a class defining variable but is associated with the risk of acquiring SAB.[19] A diagnosis of diabetes mellitus (of any severity) was consistently associated with subphenotypes D (CKD SAB) and A (older age, multimorbidity SAB; **Figure 2B**).

Bacterial genotype

The distribution of *spa*-inferred clonal complexes of isolates from the Edinburgh 2 and SABG-PCS cohorts differed substantially (**Table S6**), with only CC8 and CC15 identified in both cohorts. The distribution of clonal complexes did not differ substantially between subphenotypes within either cohort (**Figure 3**). Panton-valentine leucocidin toxin gene carriage was uncommon amongst isolates from the Edinburgh 2 cohort (7 of 463 isolates, 1.5%) and did not differ substantially across the subphenotypes (**Figure S3A**). The USA300 clone accounted for 14.7% (111/755) of isolates in the SABG-PCS cohort. Although numerically more common in subphenotype E, this difference was not statistically significant ($P = 0.2$; **Figure S3B**).

Outcomes associated with clinical subphenotypes

Subphenotype A was consistently associated with highest 90-day all-cause mortality (Edinburgh 2 47.9%, IDISA 78.3%, SABG-PCS 77.9%), whereas B (21.1%, 39.4%, 20.2%) and especially E (4.0%, 11.2%, 21.2%) were associated with the lowest (**Figure 4A; Figure S4; Tables S3-S5**). Persistent bacteremia was most common in subphenotypes C (4.1%, 16.1%, 36.0%) and E (10.2%, 14.5%, 53.7%; **Figure 4B; Tables S3-S5**). Subphenotype D was consistently associated with recurrent bacteremia although this difference was not statistically significant (**Figure 4C; Tables S3-S5**).

DISCUSSION

Approaches to achieve patient stratification remain an unmet need in SAB, to facilitate personalized clinical management and to address the heterogeneity in responses to treatments investigated in clinical trials. In this study we have analyzed three cohorts of adults with SAB, including a total of 1708 patients, replicating the five clinical subphenotypes we previously described[6]. Importantly, we included geographically diverse observational studies reflecting 'real-world' patients with SAB, including varied *S. aureus* genetic lineages, and a high proportion of MRSA and USA300 bacteremia. Consistent with our original report, subphenotypes represent five clinically distinct archetypes of SAB[7], and differ in survival and microbiologic outcomes. The reproducible identification of clinically-relevant subphenotypes supports the contention that investigation of SAB as a single syndrome is overly-simplistic and may be impeding efforts to personalize care and improve outcomes.

Multimorbidity, and diabetes specifically, were more prevalent in the subphenotypes associated with highest mortality (A and D), highlighting the need to understand how these processes impact on host defense and response to treatment. Immune dysfunction associated with CKD could also be a specific contributor to the increased risk of mortality and potentially recurrence in subphenotype D. This would be consistent with the overall increased risk of infection-related mortality in people with CKD, including in comparison to other comorbidities [20-22]. Identification of metastatic foci was more prevalent in subphenotypes C (community acquired metastatic SAB) and E (injection drug use associated SAB) which were consistently associated with persistent bacteremia but not with death. The predisposition to metastatic infection in subphenotype C remains unclear, not related to the host or pathogen characteristics investigated in this study. Investigation of host responses and impact of pathogen genetic variation at the whole genome level in this subphenotype could be particularly informative to define host and/or pathogen determinants of metastatic infection during SAB.

Our study has several strengths providing confidence in our conclusions. The current analysis includes a large cohort from the USA with a high prevalence of MRSA bacteremia (and the USA300 clone), which was missing from our original study. In comparison to other variables, MRSA infection itself did not make a large impact on class assignment, providing re-assurance that the subphenotypes are generalizable to settings with a higher prevalence of MRSA. Related to this, the distribution of *S. aureus* genetic lineages was almost entirely different in the Edinburgh 2 *vs.* SABG-PCS cohorts, providing further re-assurance regarding generalizability. We studied three well characterized patient cohorts which utilized the same definitions of the class-defining variables used in the analysis. There was a very high degree of overlap in availability of the variables, and all patients had follow-up extending to ≥90 days from the date of index blood culture. Two of the three cohort studies were prospective. The use of three observational cohorts provides confidence these subphenotypes are applicable to patients encountered during 'real life' clinical practice, which can differ from patients included in clinical trials[5, 23].

Subjectivity in model selection represents an important limitation to our approach. For the Edinburgh 2 cohort, the 5-class model had the lowest BIC value and best clinical interpretability. In both the IDISA and SABG-PCS cohorts, although the BIC value marginally favored the 4-class models, clinical interpretability favored the 5-class models, with an acceptable number of patients in the smallest class and no significant differences in entropy (as a measure of class separation). In addition, the study time periods were not overlapping between cohorts, data collection was retrospective in the Edinburgh 2 cohort, and definitions of persistent SAB differed. Despite the differences in the definition of persistent SAB, consistent associations with subphenotypes were observed, supporting the robustness of these associations. The competing risk of death was not considered when examining associations between subphenotype membership and persistent or recurrent SAB. The low number of recurrences in each cohort prevented statistical confirmation of a difference between the subphenotypes. Granular definitions of sepsis (such as the Sequential Organ Failure Assessment score) were not recorded for all cohorts so could not be included in the model nor compared between subphenotypes. Future work will define a prediction model for use in clinical practice and research studies for subphenotype assignment of individual patients. This will then allow prospective evaluation of the impact of subphenotype assignment on personalizing diagnostic and therapeutic strategies in clinical trials.

An alternative approach has recently been reported by Bélen Gutiérrez-Gutiérrez and colleagues[24], with a focus on early clinical risk stratification using variables available within the first 24 hours of the index blood culture. This approach identified three clusters based on portal of entry (skin or soft tissue; intra-vascular, mainly catheter-related; and other/unknown) with subsequent identification of clusters within each with higher or lower risk of 30-day mortality. We agree this is a useful and carefully validated tool for clinicians seeking to stratify the risk of death early in infection. This differs from and complements our approach which has defined holistic subphenotypes of disease which differ in mortality, and also microbiologic outcomes and response to adjunctive rifampicin (in the ARREST trial).

In summary, we report five reproducible clinical subphenotypes of SAB which provide a framework to rationalize the substantial heterogeneity intrinsic to this high morbidity and mortality infection. The current study extends the total number of patients included in our analyses to 3138, spanning five separate studies from the UK, Spain, the Netherlands, and the USA. We have additionally established the relevance of these subphenotypes in the setting of MRSA bacteraemia, the USA300 clone, and patients with a spectrum of multimorbidity. We think these subphenotypes are robust and can be used to stratify recruitment to and analysis of clinical trials in SAB, in addition to investigation of disease pathogenesis.

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Potential conflicts of interest

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Author contributions

Conceptualization: CDR and MCS. Data curation: CDR, MCS, ACW, FR, RK, JJ, PL, LT, TWV. Formal analysis: MCS, ZB, CDR. Investigation: MCS, ZB, ACW, CDR. Project administration: CDR, FR, MCS, ACW. Software: MCS, ZB. Supervision: CDR and MCS. Validation: CDR and MCS. Writing – original draft: MCS and CDR. Writing – review and editing: MCS, ZB, ACW, RK, JJ, PL, LT, SD, RKS, MGJB, GHG, DHD, TWV, VGF, CDR.

Data sharing

Please contact the corresponding author to discuss. We would welcome opportunities to share data and contribute to collaborative analyses.

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The idisa study team

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TABLE

Table 1: Comparison of cohort characteristics

Characteristic	Edinburgh 2 n=463	IDISA n=490	SABG-PCS n=755	*P* Value
Location	NHS Lothian, Scotland, UK	Randstad region, The Netherlands	Duke University Medical Center, North Carolina, USA	–
Time period	August 2022–December 2024	July 2017–September 2019	January 2016–January 2020	–
Age, years[1]	65 (52-79)	68 (57-77)	61 (49-71)	<0.001
Sex, male[1]	302 (65.2)	327 (66.7)	470 (62.3)	0.2
Acquisition[1]				<0.001
Community-acquired nonhealthcare-associated	199 (43.0)	166 (33.9)	322 (42.6)	
Community-acquired healthcare-associated	120 (25.9)	163 (33.3)	333 (44.1)	
Nosocomial	144 (31.1)	161 (32.9)	100 (13.2)	
Comorbidities				
Charlson Comorbidity Index	4 (2-6)	5 (3-8)	9 (5-11)	<0.001
Diabetes mellitus[2]	131 (28.3)	156 (31.8)	293 (38.9)	<0.001
Vascular disease[1,3]	121 (26.1)	211 (43.1)	243 (32.2)	<0.001
Chronic kidney disease[1]	32 (6.9)	135 (27.6)	145 (19.2)	<0.001
Prosthetic cardiac material[1,4]	40 (8.6)	54 (11.0)	102 (13.5)	0.03
Injection drug use[1]	49 (10.6)	5 (1.0)	83 (11.1)	<0.001
Liver disease[1]	77 (16.6)	26 (5.3)	97 (12.8)	<0.001
Dementia[1]	32 (6.9)	19 (3.9)	77 (10.2)	<0.001
Vital signs				
Heart rate, beats/min[1]	101 (89 to 114)	105 (94-120)	*See table S1*	<0.001
Temperature, °C[1]	38.4 (37.8 to 39.0)	38.9 (38.3 to 39.5)	*See table S1*	<0.001
Laboratory				
Hemoglobin, g/L[1]	115.0 (98.0-130.0)	112.8 (97.5-128.9)	NA	0.9
Creatinine, μM[1]	92.5 (67.5-144.0)	107.0 (76.0-177.5)	*See table S1*	<0.001
C-reactive protein, mg/L[1]	140.0 (54.0-240.8)	155.0 (59.0-262.0)	NA	0.2
Bacteremia characteristics				
MRSA[1]	13 (2.8)	10 (2.0)	303 (40.2)	<0.001
Any metastatic foci[1]	140 (30.2)	210 (42.9)	380 (50.3)	<0.001
Infective endocarditis	38 (8.2)	90 (18.4)	172 (22.8)	<0.001

Portal of entry[1]				<0.001
Unknown	141 (30.5)	158 (32.2)	207 (27.4)	
Intravenous catheter	91 (19.7)	108 (22.0)	74 (9.8)	
Skin or soft tissue infection	101 (21.8)	118 (24.1)	132 (17.5)	
Injection drug use	49 (10.6)	5 (1.0)	51 (6.8)	
Other	28 (6.0)	35 (7.1)	236 (31.3)	
Respiratory	23 (5.0)	34 (6.9)	84 (11.1)	
Urinary tract	30 (6.5)	32 (6.5)	22 (2.9)	
Outcomes				
Persistent bacteremia	17 (3.7)	40 (8.2)	207 (27.4)	<0.001
Recurrent bacteremia	12 (2.6)	12 (2.5)	29 (3.9)	0.3
90-day mortality	114 (24.6)	161 (32.9)	221 (29.3)	0.006

[1]class-defining variable included in latent class analysis.

[2]diabetes mellitus of any severity. Data available for n=754 in SABG-PCS dataset.

[3]peripheral vascular disease, myocardial infarction or stroke.

[4]implanted cardiac devices, including pacemakers and implantable automatic cardioverter-defibrillator and Left Ventricular Assist Devices, but not including prosthetic heart valves.

Continuous values are shown as median (interquartile range), compared using the Mann-Whitney U test (two cohorts) or the Kruskal-Wallis test (three cohorts). Categorical variables are shown as count (%), compared using the Chi-squared test. Variables not available are represented as NA. Vital signs and laboratory measurements were recorded at the time of the index blood culture (see further details in Methods). Abbreviations: NHS, National Health Service; IDISA, Improved Diagnostic Strategies in *Staphylococcus aureus* bacteremia study; SABG-PCS, *Staphylococcus aureus* Bacteremia Group - Prospective Cohort Study; MRSA, methicillin-resistant *Staphylococcus aureus.*

FIGURE LEGENDS

Figure 1: Reproducible clinical subphenotypes of *S. aureus* bacteremia

Comparison of class sizes and class-defining variables between subphenotypes for **(A)** the UK Edinburgh 2 cohort, **(B)** the Dutch IDISA cohort, and **(C)** the USA SABG-PCS cohort. Letters A to E refer to the subphenotypes described in the text. Vertical bars show the proportion of patients assigned to each subphenotype within each cohort. Heatmap cells are shaded according to row z score (ie, comparing subphenotypes) except "Acquisition" and "Source of SAB," where shading is by column z score (ie, comparing within each subphenotype). Intensity of red shading reflects a more positive z score (ie, above the mean), and intensity of blue shading reflects a more negative z score (ie, below the mean). Abbreviations: IDISA, Improved Diagnostic Strategies in *Staphylococcus aureus* bacteremia study; SABG-PCS, *Staphylococcus aureus* Bacteremia Group - Prospective Cohort Study; MRSA, methicillin-resistant *Staphylococcus aureus*; IV, intravenous; SSTI, skin or soft tissue infection; IDU, injection drug use.

ALT TEXT (Figure 1): heatmaps illustrating differences in class defining variables between subphenotypes in each cohort.

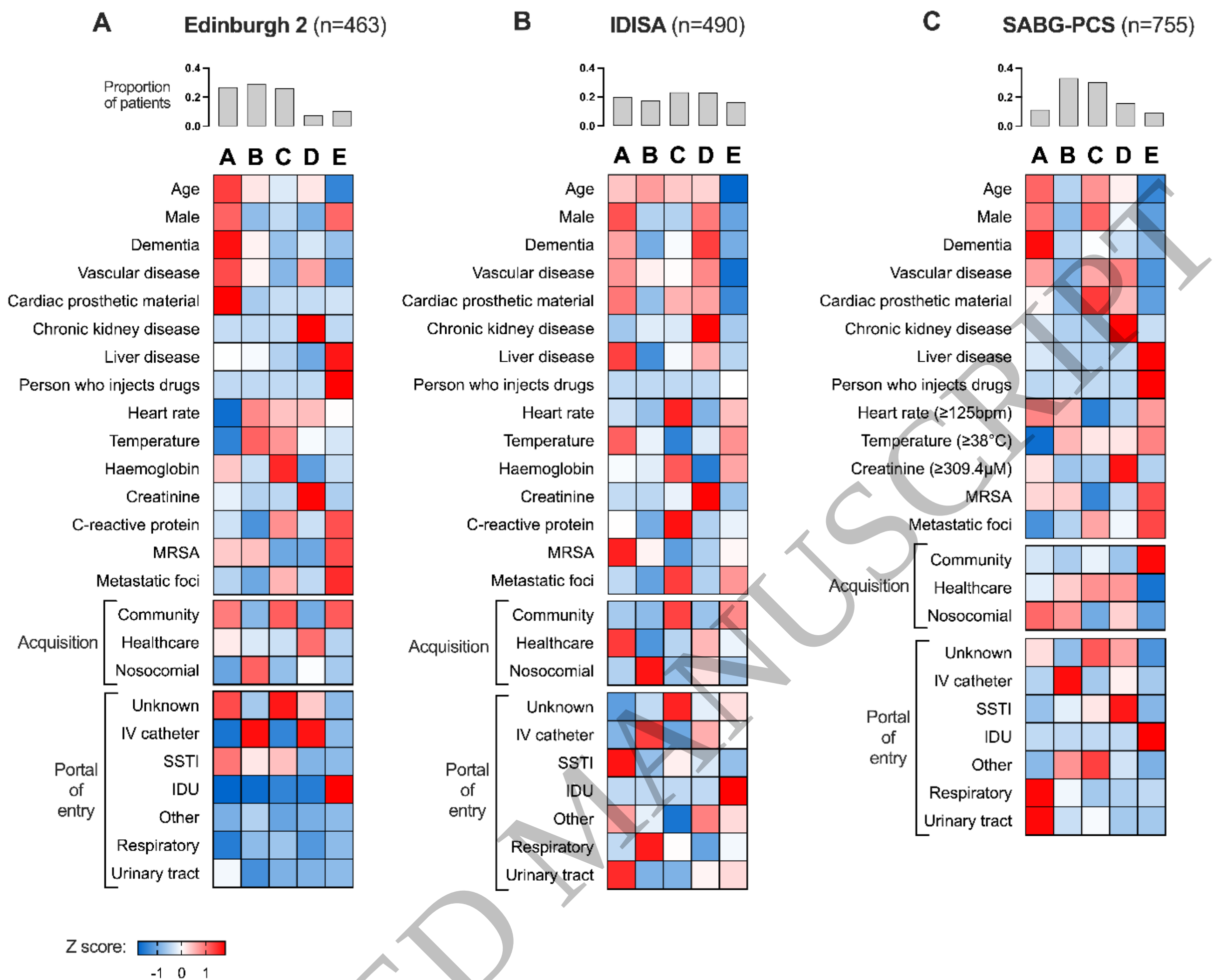


Figure 2: Multimorbidity and diabetes mellitus stratified by clinical subphenotype

Comparison of multimorbidity measured by Charlson Comorbidity Index (CCI) and diabetes mellitus of any severity between subphenotypes. **(A)** Box and whisker plot drawn using Tukey's method. Box shows interquartile range and horizontal line shows median. CCI compared between subphenotypes within each cohort using the Kruskal-Wallis test. **(C)** Bars represent z scores comparing subphenotypes within each cohort. The raw proportion of patients with diabetes mellitus was compared between subphenotypes within each cohort using Fisher's exact test. Abbreviations: IDISA, Improved Diagnostic Strategies in *Staphylococcus aureus* bacteremia study; SABG-PCS, *Staphylococcus aureus* Bacteremia Group - Prospective Cohort Study.

ALT TEXT (Figure 2): box and whisker plot showing differences in Charlson Comorbidity Index between the five subphenotypes in all three cohorts (A) and bar chart showing Z-scores for proportion of patients with diabetes mellitus in the five subphenotypes in all three cohorts.

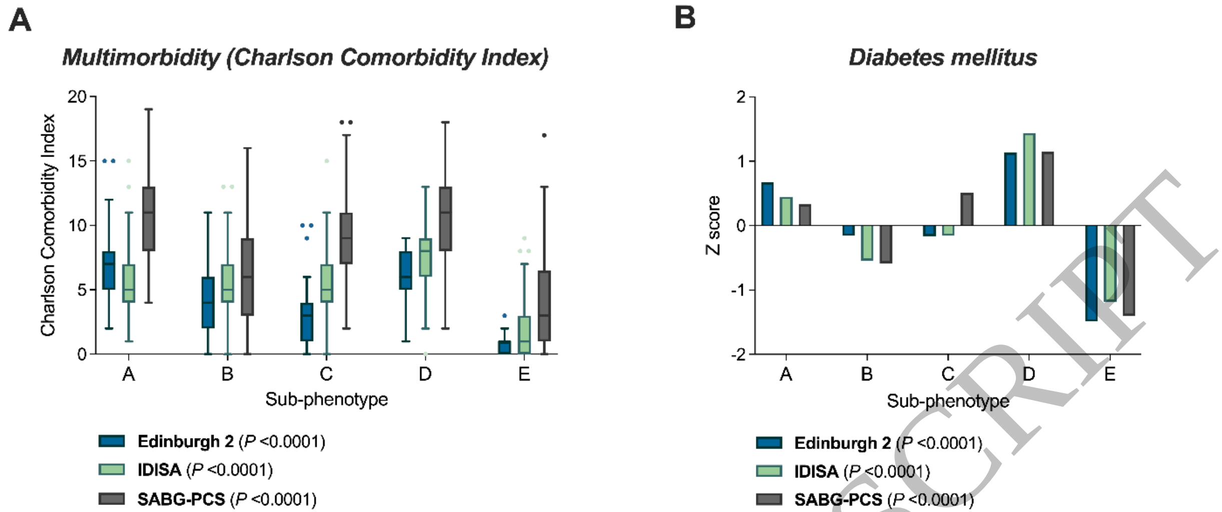


Figure 3: *S. aureus spa*-inferred clonal complex stratified by subphenotype

Spa-inferred clonal complex of *S. aureus* bloodstream isolates from **(A)** the Edinburgh 2 cohort and **(B)** the SABG-PCS cohort, stratified by subphenotype. The proportion relates to the proportion of isolates within each subphenotype. Abbreviations: SABG-PCS, *Staphylococcus aureus* Bacteremia Group - Prospective Cohort Study; CC: clonal complex; NA: not available.

ALT TEXT (Figure 3): stacked bar charts showing distribution of spa-inferred clonal complexes in each subphenotype in (A) the Edinburgh 2 cohort and (B) the SABG-PCS cohort.

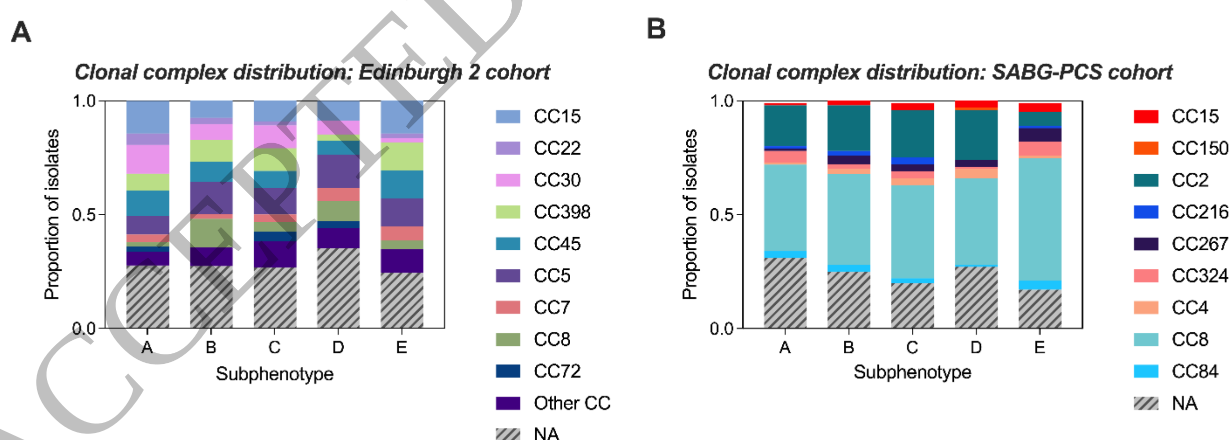
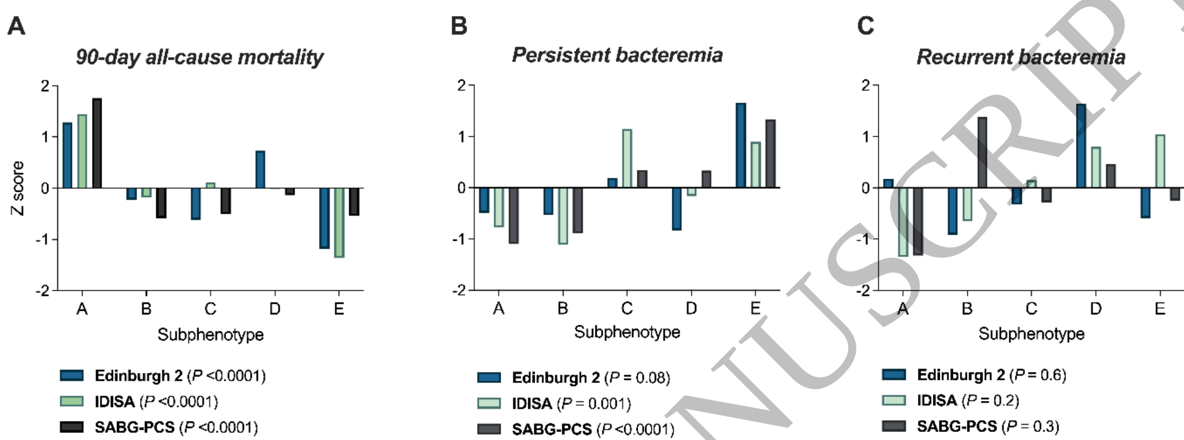


Figure 4: Survival and microbiologic outcomes differ between clinical subphenotypes

Comparison of **(A)** 90-day all-cause mortality, **(B)** persistent bacteremia, and **(C)** recurrent bacteremia between subphenotypes. Bars represent z scores comparing subphenotypes within each cohort. The raw proportion of each outcome was compared between subphenotypes within each cohort using Fisher's exact test. Abbreviations: IDISA, Improved Diagnostic Strategies in

Staphylococcus aureus bacteremia study; SABG-PCS, *Staphylococcus aureus* Bacteremia Group - Prospective Cohort Study.

ALT TEXT (Figure 4): bar charts showing Z-scores for proportion of patients with (A) all-cause mortality by 90 days, (B) persistent bacteremia, and (C) recurrent bacteremia, in the five subphenotypes in all three cohorts.



For Prescribing Information and Adverse Event Reporting please see bottom of email
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Rukobia▼ (fostemsavir)

Managing multidrug-resistant HIV-1 presents unique challenges[1]

Rukobia, in combination with other antiretrovirals, is indicated for the treatment of adults with multidrug resistant HIV-1 infection for whom it is otherwise not possible to construct a suppressive antiviral regimen

Rukobia▼(fostemsavir) provides an option to be added in or used as a replacement ARV in people who are experiencing difficulty achieving viral suppression on their current ART. Here, you'll learn how Rukobia works differently from other ARVs and about its durable efficacy and robust CD4+ T-cell outcomes from the BRIGHTE trial over 5 years. Are you ready to see how Rukobia could help you meet the needs of eligible people living with MDR HIV-1?

Discover what's possible with RUKOBIA:

Leads to a promotional ViiV website

- **Durable virologic suppression, demonstrated over 5 years of follow-up in the BRIGHTE Study.[3]**
- **Robust CD4+ T-cell recovery, as per the BRIGHTE study.[3]**
- **Low potential for clinically significant DDIs, as RUKOBIA can be co-administered with most common treatments for HIV and frequently prescribed medicines for comorbidities[2]**
- **Consistent safety and tolerability profile, with 5 years of follow-up data, as per the BRIGHTE study[2,3]**

Adverse events should be reported. Reporting forms and information can be found at https://yellowcard.mhra.gov.uk/ or search for MHRA Yellowcard in the Google Play or Apple App store. Adverse events should also be reported to GSK via the GSK Reporting Tool or on 0800 221441.
Prescribing Information

ART, antiretroviral therapy; **ARV,** antiretroviral; **CD4,** cluster of differentiation 4; **DDI,** drug–drug interaction; **HIV-1,** human immunodeficiency virus type 1; **MDR,** multidrug-resistant.

References

1. Ackerman P, Thompson M, Molina JM, et al. Long-term efficacy and safety of fostemsavir among subgroups of heavily treatment-experienced adults with HIV-1. AIDS. 2021;35(7):1061-1072.
2. RUKOBIA Summary of Product Characteristics.
3. Aberg J et al. Infect Dis Ther 2023; 12:2321-2335.

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